Exercises Week 4

Raul Pekander

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Exercise 1

Let $X(t)$ be a Gaussian zero-mean process with t $\in \mathbf{R}^d$, d = 1,2, and with Matérn covariance function.

- Let's draw a realization of one-dimensional Matérn with $\nu = 3/2$

  Let's use the following discretisation

  h = 0.02, process is [0,10], N = 501. Every fifth node is observed
  The Matérn covariance 3/2

$$C(r) = \sigma_f^2 \left(1 + \frac{\sqrt{3r}}{l}\right) \exp\left(-\frac{\sqrt{3r}l}{l}\right) \tag{1}$$

  (Rasmussen, 2005) with values $l = 0.4$,$\sigma_f = 1$ and noise $\sigma_n = 0.1$.

  The prior for the correlation length: $\log l \sim$ N(log 0.5, 0.7^2). Its value is updated with a random walk Metropolis-Hasting step on log l with step size of 0.15.

  Now the full covariance model has a prior $x \sim N(0, C_l)$. C_l represents a dense Matérn matrix. With the Cholesky factor $C_l = LL^T$, the prior density of is proportional to :

$$\det(L)^{-1} \exp(-\frac{1}{2}||L^{-1}x||^2) \tag{2}$$

  The computational cost for each update of l is $O(N^3)$.

  So we propose a stochastic partial differential approximation in d=1.

$$\begin{aligned}(\kappa^2 - \frac{d^2}{dt^2})x(t) &= W.N \\ \kappa &= \frac{\sqrt{3}}{l}\end{aligned} \tag{3}$$

Once we consider finite difference and Diriclet boundary conditions, we have tridiagonal matrix L where

$$
\begin{aligned}
a_0 &= \kappa^2 + 2/h^2 \\
a_1 &= -1/h^2 \\
L &= \begin{bmatrix} a_0 & a_1 & 0 & \dots & 0 \\ a_1 & a_0 & a_1 & \dots & 0 \\ 0 & a_1 & a_0 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & a_1 \\ 0 & \dots & \dots & a_1 & a_0 \end{bmatrix}
\end{aligned} \tag{4}
$$

The prior is $x = sL^{-1}z, z \sim N(0, I)$. The scale factor is denoted by $s = c\kappa^{3/2}$. So then the precision matrix is $L^T L/s^2$. The constant c is calibrated once at $l = 0.4$. The matrix is sparse and its computational cost is $O(N)$.

A normalisation constant is used. The prior density of the SPDE model is

$$p(x|l) = (2\pi)^{-N/2} \frac{\det(L)}{s^N} \exp(-\frac{||Lx||^2}{2s^2} \tag{5}$$

and the close-form determinant is

$$\lambda_k = a_0 - \frac{2}{h^2} \cos\left(\frac{k\pi}{N+1}\right) \tag{6}$$

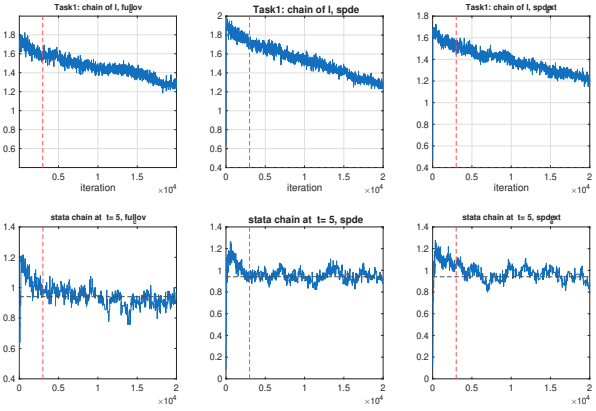
$$\log\det L = \sum_{k=1}^{N} \log \lambda_k \tag{7}$$

The computational cost is $O(N)$. (Lindgren et al., 2011).

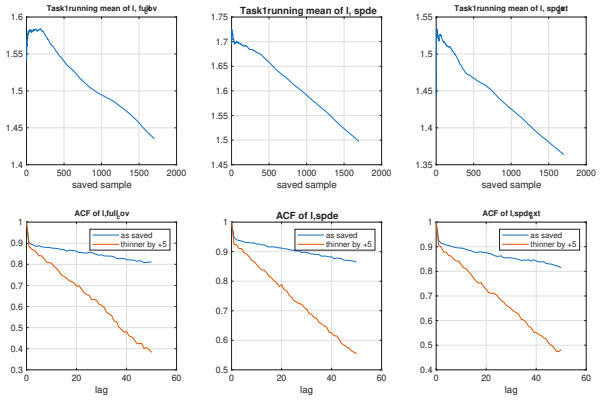
To solve the boundary behaviour problem, a dirichlet condition is used where the ends of the field are pushed to zero. This reduces the prior variance near the boundaries. Domain extension is to alleviate the effect by extending the domain by $2 \cdot l_{true}/h$ nodes on each side. Only inner nodes observations are kept and reported.
To sample we use pCn to propose a new state: $u' = \sqrt{1-\beta^2}u + \beta\xi, \xi \sim N(0, C_l)$. The acceptance happens with probability: $\min\{1, \exp(\Phi(u) - \Phi(u')\}$, where:

$$\Phi(u) = \frac{||y - Hu||^2}{2\sigma_n^2} \tag{8}$$

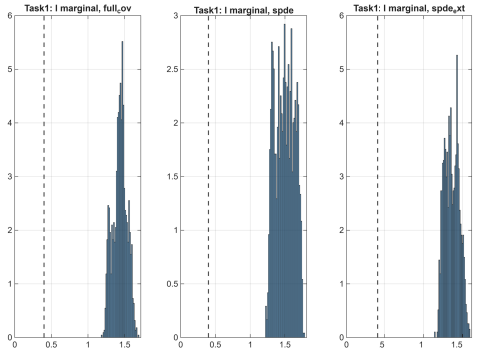
, H being the observation matrix. In this case we use $\beta = 0.15$, $n_{mcmc} = 20000$, burn-in 3000, and thinning factor of 10.



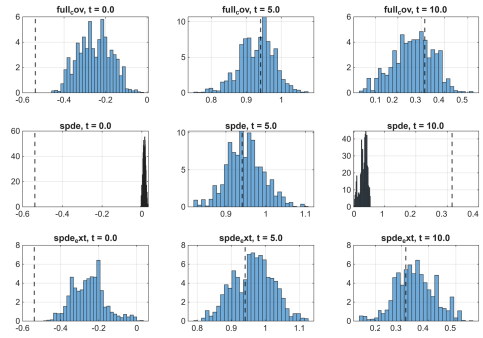
(a) State and l chains



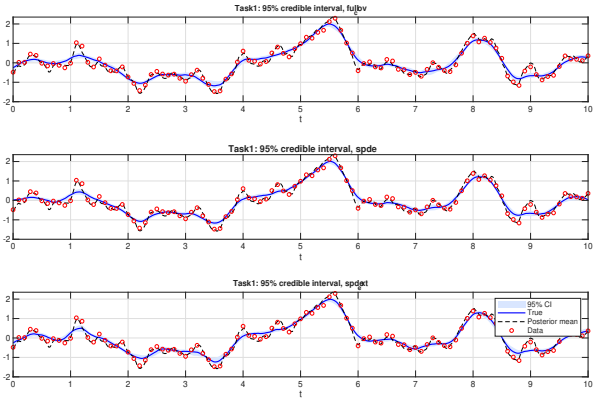
(b) Task 1.1 MCMC



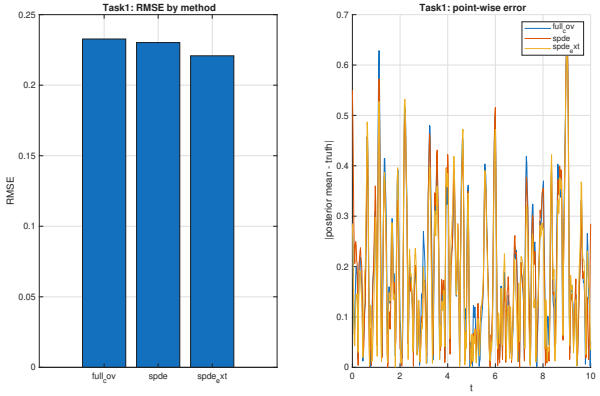
(c) Task 1.1 posterior



(d) Task 1.1 RMSE



(e) Task 2.1 results



(f) Task 2.1 MCMC

In Figure (a) that every method the parameter l downward drift from 1.7-2.0 to 1.2 1-3. This could indicate that the burn-in period is too short and there isn't an established stable posterior distribution. The difference between the methods is small in this project. The state, at t=5, does however converge rapidly and the fluctuation is within the desired range.

Figure (b) shows the running mean and ACF. For the parameter l, the method don't distinguish much from each other. The trend that hasn't not settled to horizontally indicate that the mean is heavily dependent on the end of simulation and is not representative of the true posterior mean. The ACF-plots show that there is

still strong auto-correlation even after 50 lags. The thinning does reduce the auto-correlation to the point that at lag 50, the ACF is between 0.40 and 0.55.

In Figure (c) in the case of $full_{cov}$ and $spde_{ext}$,we see dense distributions which imply that the posterior variance is small. The distribution of *spde* show signs of slow mixing and greater uncertainty. All the methods overestimate the parameter l. A larger number of iterations could bring the distributions closer to the true value. The extended SPDE model has very similar shape than the full covariance model. It is an indication that it can approximately the Gaussian process well since there is no distortion of the boundary effects.

Figure (d) shows that at t=5, all of the methods perform well. The peaks of the distributionsconcentrate around the true value. At the boundaries, t=0 and t=10.0, the ordinary SPDe model produces distribution that are far off from the true value, narrow and concentrated around 0. The extended SPDE model has a distribution much closer to the full covariance model: both have wide distribution and on t=10 the distribution overlaps the true value.

Figure (e) shows that all three methods are able to track the signal accurately.

In Figure (f) we see that the extended SPDE gave the lowest RMSE. The point-wise RMSE tells that the methods share same peaks for the most part and the residuals cannot be simply explained by the choice of method but rather by noise and observation frequency.

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Gain full_cov/SPDE: 96.6x
      Method        RMSE       Coverage      E_l
    __________    _______    ________    ______

    "full_cov"    0.23274    0.33333     1.4352
    "spde"        0.23019    0.32335     1.4973
    "spde_ext"    0.22085     0.3513     1.3635
```

Figure 2: Performance of the methods

In Figure 2, we confirm the fact that the computation of SPDE methods is much faster compared full-covariance matrix computation. This can be explained by the fact that SPDE transform the dense covariance matrices into sparse precision matrices. The accuracy doesn't suffer from speed and in the case of SPDE where we treat the boundary effects, the RMSE is the smallest in comparison to other methods. The extended SPDE method also has the best coverage and its expected value for parameter l is the closest to the true value of 1.50.

References

Lindgren, F., Rue, H., and Lindström, J. (2011). explicit link between gaussian fields and gaussian markov random fields: the stochastic partial differential equation ap-

proach. *Journal of the Royal Statistical Society. Series B, Statistical methodology*, 73(4):423–498.

Rasmussen, Carl Edward, W. C. K. (2005). *Gaussian Processes for Machine Learning*. MIT Press.